Step 1:

Here, more length means more days. So, it is in direct proportion.

$\therefore$ The multiplying factor is $\frac{512}{180}$ (take the reciprocal).

Step 2:

Also, more women means less days. So, it is in inverse proportion.

$\therefore$ The multiplying factor is $\frac{15}{32}$ (no change).

Step 3:

$\therefore x = 12 \times \frac{512}{180} \times \frac{15}{32} = 16$ days.

Formula Method:

Here, $P_1 = 15$, $D_1 = 12$ and $W_1 = 180$ and $P_2 = 32$, $D_2 = x$ and $W_2 = 512$

Using the formula, $\frac{P_1 \times D_1}{W_1} = \frac{P_2 \times D_2}{W_2}$

$$\Rightarrow \frac{15 \times 12}{180} = \frac{32 \times x}{512}$$

$$\Rightarrow 1 = \frac{32 \times x}{512} \Rightarrow x = \frac{512}{32} = 16 \text{ days.}$$

Remark: Students may answer in any of the three given methods dealt here.



Try these

1. If x and y vary directly, find k when $x = y = 5$.
2. If x and y vary inversely, find the constant of proportionality when $x = 64$ and $y = 0.75$



Activity

Draw a circle of a given radius. Then, draw its radii in such a way that the angles between any two consecutive pair of radii are equal. Start drawing 3 radii and end with drawing 12 radii in the circle. List and prepare a table for the number of radii to the angle between a pair of consecutive radii and check whether they are in inverse proportion. What is the proportionality constant?

Example 4.21 **(Inverse – Direct Variation)**

If 81 students can do a painting on a wall of length 448 m in 56 days, then how many students can do the painting on a similar type of wall of length 160 m in 27 days?

Solution:

Multiplicative Factor Method

Students	Days	Length of the wall (Work)
81 I D x	56 I 27	448 D 160

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